

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
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- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern	Isomer AI	May 2025 – May 2026
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- Collaborated with product and design teams to ship 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows; 923 commits, 7 production deployments.
- Owned full-stack React/Node.js + TypeScript codebase: PostgreSQL, Redis, Docker, Auth0 RBAC, and Playwright test coverage.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
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- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
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- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
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- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
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- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

LLM Eval Harness	Independent Project	2025
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- Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks.
- Tracked hallucination rate, p50/p95 latency, and cost-per-call; applied McNemar significance tests across 6 metrics.

AlgoPulse	Full-Stack Agentic AI Platform	2025
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- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.
- Replaced polling with event-driven Supabase Edge Functions; sustained sub-200ms SSE delivery at 2,000 req/sec.

CortexQ	Kubernetes LLM Inference Platform	2026
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- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability; 88 pytest tests.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **AI & Automation:** Claude API, GPT-4o, Gemini, LangChain, RAG, Temporal.io, Kafka, Prompt Engineering, DuckDB
- **Tools & Integration:** FastAPI, Node.js, REST APIs, PostgreSQL, Redis, MongoDB, Docker, GitHub Actions, GCP, AWS
- **Evaluation & Observability:** LLM Evals, OpenTelemetry, Prometheus, Grafana, Playwright, Testcontainers, k6

Additional Experience and Awards

- **GDSC Lead:** Led Google Developer Student Club at Mumbai University: ran 4+ hackathons and workshops reaching 500+ students; coached beginner participants into first-time model builders and follow-on event leads.
- **Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog for general and technical audiences.
- **Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- **IEEE Published:** Co-authored Computer Vision-Based Cybersecurity Threat Detection research; presented at Springer icSoftComp 2023 to 500+ global attendees.